LETTER

The welfare effects of public school closures: Evidence from rural Denmark

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ABSTRACT

Using transaction-level housing data from 2006–2022, we estimate a hedonic difference-in-differences model to identify the capitalization effects of rural public school closures in Denmark. We exploit a reform in which eight schools were closed simultaneously in a sparsely populated municipality, where nearby substitutes are limited. We find that closures reduced nearby house prices by 17–24 percent, with effects persisting over time and declining with distance. The estimates imply substantial welfare losses for affected households and a high willingness to pay for continued access to a local public school. The study provides causal, welfare-interpretable evidence on the value of rural public schools as local public goods.

KEYWORDS

Hedonic house price; Difference-in-Difference; Rural amenities

JEL CLASSIFICATION

R21; H41; C21; I28

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1. Introduction

A large literature shows that school quality and access are capitalized into housing prices (Wada and Zahirovic-Herbert, 2013; Feng and Lu, 2013; Ries and Somerville, 2010), yet causal evidence on public school closures remains limited. Only two quasi-experimental studies examine school closures directly, both focusing on isolated closures in urban or metropolitan US settings. They find that the loss of a neighborhood school reduces nearby house prices by roughly 7–10 percent (Bogart and Cromwell, 2000; Rosburg et al., 2017).

These studies provide important evidence on the amenity value of local schools, but their settings offer limited insight into rural and small-town closures. In rural housing markets, nearby substitutes are fewer, and school closures are often driven by public-service consolidation rather than school performance (Sher and Tompkins, 2019). Local schools may therefore matter not only as providers of education, but also as community meeting places and anchors of residential sorting, particularly for families with children (Cedering and Wihlborg, 2020; Sørensen et al., 2021).

This letter addresses this gap by exploiting a quasi-experimental reform in which eight public schools closed simultaneously in a rural Danish municipality. Using transaction-level house price data and a hedonic difference-in-differences design, we estimate the capitalization and welfare effects of losing access to a local public school. We document a persistent decline in house prices, consistent with substantial welfare losses in affected rural communities.

2. Data and institutional setting

We use 5,905 transactions of single-family houses, apartments, and rowhouses in Tønder Municipality, Denmark, from 2006 to 2022. These data are merged with a register of all public schools operating in the municipality during the period, including schools that later closed. Before the reform, Tønder Municipality had 18 public schools, each serving a distinct town and its surrounding area. In 2010/2011, eight schools were closed simultaneously, while affected pupils were reassigned to other public schools within the municipality. The reform was motivated by declining fertility across all areas and municipal budget pressure[1].

We exploit the spatial structure of the reform to define treatment and control areas. Because pre-reform school district boundaries are unavailable, each dwelling is assigned to its nearest public school based on pre-reform distances. Dwellings whose nearest school subsequently closed are assigned to the treatment group, yielding eight treatment areas and ten control areas. This proximity-based assignment approximates the local catchment areas served by each school before the reform.

Two features of the Danish public school system are important for interpreting the results relative to the existing US evidence. First, school district boundaries are soft: households may choose another public school if seats are available. The relevant welfare loss is therefore the loss of access to a local public school, not access to public education per se. Second, per-pupil expenditures are broadly equalized within municipalities and are not tied to local property taxes.

Descriptive statistics show that treatment areas were negatively selected before the reform: average pre-2010 house prices were approximately $\$35,000$ lower than in control areas. However, Figure 1 shows that average prices in treatment and control areas followed similar trends before the school closures were announced. The treatment-control price gap widens in 2010, when the closures were announced, and persists thereafter, despite a broader decline in the regional housing market. Full descriptive statistics are reported in Appendix A.1.

[1]No weight was put on the academic performance when choosing which schools to close. However, the closed schools were smaller, on average.

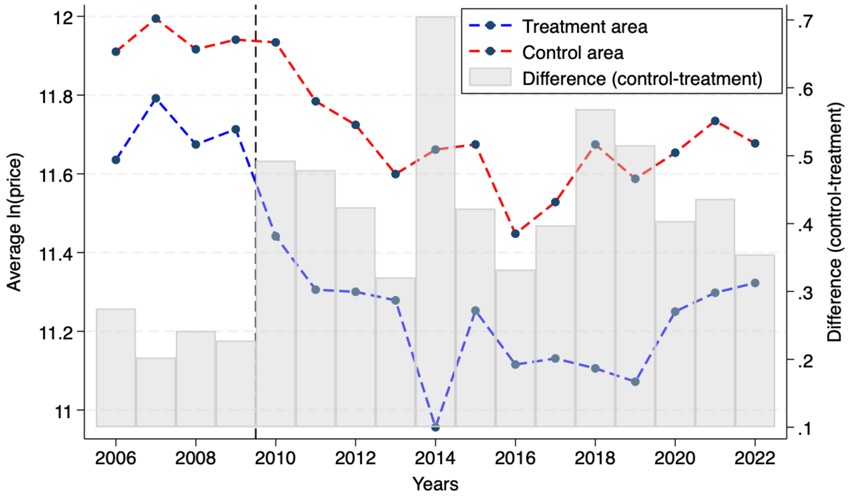

Figure 1. Average house prices in treatment and control areas

Note: The figure shows average log house prices in treatment and control areas. Gray bars show the treatment-control difference. The dotted line marks the 2010 school-closure announcement. Prices are inflation-adjusted and measured in US dollars.

3. Identification strategy

We estimate the capitalization effect of public school closures using a hedonic difference-in-differences design. The model compares changes in house prices before and after the reform between dwellings assigned to closed and non-closed schools. Since the effect is expected to vary with proximity to the closed school, we allow treatment effects to differ across distance bins:

$$\begin{aligned} p_{i,t} = \theta_1 D_{g(i),t}\mathbf{1}(dist_i < 2) + \theta_2 D_{g(i),t}\mathbf{1}(2 \leq dist_i < 4) \\ +\theta_3 D_{g(i),t}\mathbf{1}(4 \leq dist_i < 6) + \theta_4 D_{g(i),t}\mathbf{1}(dist_i \geq 6) \\ +\mathbf{Z}'_{i,t}\beta + \alpha_{g(i)} + \lambda_t + \epsilon_{i,t} \end{aligned} \tag{1}$$

Here, $p_{i,t}$ is the log transaction price of dwelling i in year t, and $D_{g(i),t}$ equals one for dwellings in treatment areas sold from 2010 and onwards. We use 2010 as the treatment year because the school closures were publicly announced that year, although some anticipation may have occurred as the political process began in 2009 (Svendsen and

Sørensen, 2016). The variable $dist_i$ measures distance in kilometers to the nearest pre-reform public school. The coefficients θ_h therefore measure the percentage change in house prices after a closure for dwellings in each distance bin.

The vector $\mathbf{Z}i, t$ includes housing and location characteristics, including distance-bin indicators. Area fixed effects, $\alpha_{g(i)}$, absorb time-invariant differences between school areas, while year fixed effects, λ_t, absorb common shocks to the housing market. The identifying assumption is that, absent the school closures, house prices in treatment and control areas would have followed parallel trends. The pre-reform price trends shown in Figure 1 support this assumption.

Following Kuminoff and Pope (2014) and Banzhaf (2021), we also estimate an extended specification that allows implicit prices of housing attributes to change after the reform by interacting housing characteristics with the post-treatment indicator. This helps distinguish welfare effects from shifts in the hedonic price function and allows the estimated capitalization effect to be interpreted as an upper bound on the welfare loss. All models are estimated by OLS with area and year fixed effects. Standard errors are clustered at the school-area level[2]; results are robust to classical robust standard errors and wild cluster bootstrap inference.

4. Results

Table 1 reports the estimated capitalization effects across distance bins. Column (1) presents the baseline specification from Equation 1. Houses located within two kilometers of a closed school experienced a price decline of approximately 24 percent. This effect is substantially larger than the 7–10 percent effects found in previous studies of isolated urban or metropolitan school closures, consistent with the rural setting, fewer nearby substitutes, and the simultaneous closure of multiple schools within the municipality.

The estimates reveal a clear spatial gradient. The effect is largest within two kilometers of a closed school and declines with distance. Houses located between two and four kilometers from a closed school experienced a price decline of approximately 20

[2] We are aware that this leaves us with only 18 clusters.

percent, while estimates for houses located more than four kilometers away are smaller and statistically insignificant. This pattern indicates that the housing-market response is concentrated close to the closed schools and diminishes as the loss of local access becomes less salient.

Column (2) reports the extended specification, which allows implicit prices of housing attributes to change after the reform. Under this specification, the estimated decline for houses within two kilometers of a closed school falls from 24 to 17 percent, but remains statistically significant. The estimate for houses located between two and four kilometers falls from 20 to 14 percent and becomes statistically insignificant, while effects beyond four kilometers are close to zero. The difference between columns (1) and (2) suggests that part of the baseline capitalization effect reflects changes in the hedonic price function after the reform. Nevertheless, both specifications point to a substantial and spatially concentrated welfare loss for households closest to the closed schools.

The positive and marginally significant estimate for dwellings located more than six kilometers from a closed school in column (2) does not appear in the baseline specification and should therefore be interpreted with caution. Additional analyses provide no evidence that closures redirected housing demand toward unaffected parts of Tønder Municipality. In a dynamic DiD comparing the control areas of Tønder with neighboring municipalities, house prices moved in parallel, suggesting no detectable spillover effects. Year-specific treatment estimates also show that the treatment-control price gap emerges in 2010, when the closures were announced, rather than during the financial crisis in 2008–2009. Finally, the statistical significance of our results is not driven by the use of clustered standard errors, although we have only 18 clusters. The results are robust to alternative inference procedures, including classical robust standard errors and wild cluster bootstrap inference.

Table 1. Empirical Results from a heterogeneous effects model

Variable	(1)	(2)
$D_{g(i),t}\mathbf{1}(dist_i < 2km)$	-0.24***	-0.17***
	(0.04)	(0.04)
$D_{g(i),t}\mathbf{1}(2km \leq dist_i < 4km)$	-0.20*	-0.14
	(0.09)	(0.08)
$D_{g(i),t}\mathbf{1}(4km \leq dist_i < 6km)$	-0.07	-0.00
	(0.14)	(0.13)
$D_{g(i),t}\mathbf{1}(dist_i \geq 6km)$	0.01	0.08*
	(0.05)	(0.05)
Control Var	X	X
Area FE	X	X
Year FE	X	X
Changing implicit prices		X
Observations	5,905	5,905
R-squared	0.45	0.46

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Clustered standard errors in parentheses. $dist_i$ denotes distance to the nearest pre-reform public school.

5. Conclusion

This letter shows that simultaneous rural public school closures generate substantial and persistent declines in local housing values. The closure of eight public schools reduced house prices by 17–24 percent in exposed areas. Interpreted through a hedonic framework, this corresponds to a willingness to pay of roughly $25,000$ per household for access to a local public school. Aggregating across affected properties suggests a total loss in local housing wealth of approximately 37.4 million.[3]

The results indicate that local public schools are important amenities in rural housing markets. Beyond providing education, schools serve as focal points for community life and influence residential sorting, particularly for families with children. This helps explain why the estimated capitalization effects are larger than those found in less rural settings.

[3] As of 2022, 1,496 dwellings are located within two kilometers of a closed public school. The total loss in housing wealth is therefore: $Welfare = Avg.\ WTP \times No.\ affected\ houses$ =25,000×$1,496$ =37,400,000.

The findings highlight an important trade-off for local governments. While school consolidation may generate fiscal savings, it can also impose sizable and persistent welfare costs on local residents. By exploiting a large rural consolidation reform, the paper provides causal, welfare-interpretable evidence on the value of local public schools as public goods.

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Appendix A. Tables

A.1. Descriptive statistics

Table A1. Means table before/after 2010 across treatment and control areas

Variable	Treatment area		Control Area	
	Before 2010	After 2010	Before 2010	after 2010
Price (Dollars)	140,075	97,507	175,010	142,728
Year	2007	2016	2007	2016
Single family house (fraction)	0.96	0.98	0.92	0.94
Apartment (fraction)	0.02	0.01	0.03	0.03
Rowhouse (fraction)	0.02	0.01	0.05	0.03
Size (m2)	150.45	153.47	146.95	150.05
Size squared (m2)	26143.46	27102.44	24611.35	25200.98
Number of bathrooms	1.19	1.23	1.26	1.29
Number of toilets	1.36	1.42	1.52	1.56
Number of rooms	4.78	4.89	4.77	4.82
Number of floors	1.01	1.01	1.04	1.04
Brick (fraction)	0.96	0.96	0.95	0.96
Wood (fraction)	0.02	0.01	0.01	0.01
Lightweight concrete (fraction)	0.02	0.02	0.03	0.02
Concrete or timbered (fraction)	0.00	0.00	0.00	0.01
Outhouse (fraction)	0.15	0.15	0.16	0.16
Car park (fraction)	0.03	0.04	0.06	0.09
Double car park (fraction)	0.02	0.02	0.04	0.05
Roof of fibercement asbestos (fraction)	0.51	0.51	0.44	0.45
Roof tile (fraction)	0.15	0.15	0.21	0.22
Roof other material (fraction)	0.34	0.34	0.35	0.33
Build 1477-1939 (fraction)	0.49	0.48	0.32	0.30
Build 1940-1959 (fraction)	0.14	0.17	0.19	0.14
Build 1960-1989 (fraction)	0.34	0.32	0.44	0.49
Build 1990-2009 (fraction)	0.03	0.03	0.05	0.05
Build 2010-2022 (fraction)	0.00	0.00	0.00	0.02
Oven heating (fraction)	0.01	0.01	0.00	0.00
Stove heating (fraction)	0.01	0.01	0.00	0.00
Heatpump heating (fraction)	0.18	0.20	0.08	0.08
Electric heating (fraction)	0.19	0.19	0.12	0.11
Central heating (fraction)	0.57	0.53	0.31	0.28
District heating (fraction)	0.05	0.06	0.49	0.53
Last major renovation in the 1970's (fraction)	0.09	0.09	0.09	0.10
Last major renovation in the 1980's (fraction)	0.05	0.04	0.04	0.04
Last major renovation in the 1990's (fraction)	0.10	0.09	0.08	0.08
Last major renovation in the 2000's (fraction)	0.03	0.05	0.05	0.04
Last major renovation in the 2010's (fraction)	0.00	0.01	0.00	0.01
Nearest town (size in hectares)	19.00	13.04	271.53	290.75
Urban diversity	1.34	1.34	12.35	12.99
Proximity to forrest (capped)	249.24	251.96	245.94	252.36
Proximity to lake (capped)	55.81	52.21	144.96	143.02
Proximity to wetland (capped)	44.83	48.51	68.74	66.92
Proximity to coast (capped)	1.93	7.00	0.00	0.13
Proximity to main road (capped)	53.68	55.59	35.27	30.91
Proximity to train station (capped)	65.77	50.64	151.65	155.35
Proximity to railway (capped)	6.15	7.31	6.36	6.36
Proximity to windmill (capped)	12.10	18.31	6.76	6.58

Note: The table reports summary statistics for Tønder municipality at the dwelling level. The price is inflation adjusted using the danish consumer price index. All proximity variables are constructed and capped according to Lautrup et al. (2023).

A.2. DiD model with heterogeneous effects - Full model

Table A2. Empirical Results from a heterogeneous effects model

Variable	(1)	(2)
$D_{g(i),t}\mathbf{1}(dist_i < 2km)$	-0.24***	-0.17***
	(0.04)	(0.04)
$D_{g(i),t}\mathbf{1}(2km \leq dist_i < 4km)$	-0.20*	-0.14
	(0.09)	(0.08)
$D_{g(i),t}\mathbf{1}(4km \leq dist_i < 6km)$	-0.07	-0.00
	(0.14)	(0.13)
$D_{g(i),t}\mathbf{1}(dist_i \geq 6km)$	0.01	0.08*
	(0.05)	(0.05)
$\mathbf{1}(dist_i < 2km)$	0.15**	0.15***
	(0.06)	(0.05)
$\mathbf{1}(2km \leq dist_i < 4km)$	0.15**	0.16***
	(0.05)	(0.04)
$\mathbf{1}(4km \leq dist_i < 6km)$	0.16**	0.15**
	(0.07)	(0.06)
$\mathbf{1}(dist_i \geq 6km)$	-	-
Apartment (dummy)	0.08	0.25***
	(0.07)	(0.06)
Rowhouse (dummy)	-0.05	-0.13
	(0.06)	(0.09)
Size (m2)	0.01***	0.00***
	(0.00)	(0.00)
Size squared	-0.00***	-0.00**
	(0.00)	(0.00)
Number of bathrooms	0.03	-0.02
	(0.02)	(0.05)
Number of toilets	0.04**	0.04
	(0.02)	(0.03)
Number of rooms	0.01	0.02**
	(0.01)	(0.01)
Number of floors	0.01	0.03
	(0.08)	(0.12)
Wood (dummy)	-0.07	-0.34
	(0.12)	(0.22)
Lightweight concrete (dummy)	-0.20***	-0.24***
	(0.04)	(0.07)
Outhouse (dummy)	0.02	0.01
	(0.02)	(0.03)
Car park (dummy)	-0.04	-0.10
	(0.03)	(0.15)
Car park double (dummy)	0.06	0.02
	(0.05)	(0.19)
Roof tile (dummy)	0.08***	0.05
	(0.03)	(0.03)
Roof fibercement asbestos (dummy)	-0.09***	-0.03
	(0.02)	(0.03)

Table A2. Empirical Results from a heterogeneous effects model (continued)

Variable	(1)	(2)
Build 1940-1959 (dummy)	0.10*** (0.03)	0.10*** (0.03)
Build 1960-1989 (dummy)	0.28*** (0.03)	0.28*** (0.03)
Build 1990-2009 (dummy)	0.59*** (0.07)	0.60*** (0.07)
Build 2010-2022 (dummy)	0.02 (0.23)	-0.08 (0.21)
Stove heating (dummy)	-0.36 (0.32)	-0.11 (0.21)
Heatpump heating (dummy)	-0.29 (0.27)	-0.22 (0.16)
Electric heating (dummy)	-0.32 (0.28)	-0.16 (0.14)
Central heating (dummy)	-0.33 (0.28)	-0.16 (0.14)
District heating (dummy)	-0.33 (0.27)	-0.16 (0.15)
Last major renovation in the 1970's (dummy)	0.01 (0.02)	0.00 (0.02)
Last major renovation in the 1980's (dummy)	0.13*** (0.03)	0.12*** (0.03)
Last major renovation in the 1990's (dummy)	0.23*** (0.03)	0.23*** (0.03)
Last major renovation in the 2000's (dummy)	0.30*** (0.03)	0.30*** (0.03)
Last major renovation in the 2010's (dummy)	0.36*** (0.08)	0.34*** (0.08)
Nearest town (size in hectares)	0.00*** (0.00)	0.00 (0.00)
Urban diversity	0.00 (0.00)	0.01** (0.00)
Proximity to forest	-0.00 (0.00)	-0.00 (0.00)
Proximity to lake	0.00 (0.00)	0.00 (0.00)
Proximity to wetland	0.00 (0.00)	0.00* (0.00)
Proximity to coast	0.00*** (0.00)	0.00*** (0.00)
Proximity to main road	-0.00** (0.00)	-0.00 (0.00)
Proximity to train station	-0.00 (0.00)	-0.00 (0.00)
Proximity to national railway	0.00 (0.00)	0.00 (0.00)
Proximity to windmill	-0.00 (0.00)	-0.00 (0.00)
Apartment $\times$ $1(t \geq 2010)$		-0.25*** (0.07)
Rowhouse $\times$ $1(t \geq 2010)$		0.13 (0.09)
Size $\times$ $1(t \geq 2010)$		0.00*** (0.00)
Size squared $\times$ $1(t \geq 2010)$		-0.00*** (0.00)
Number of bathrooms $\times$ $1(t \geq 2010)$		0.07 (0.05)

Table A2. Empirical Results from a heterogeneous effects model (continued)

Variable	(1)	(2)
Number of toilets × $1(t \geq 2010)$		0.00
		(0.03)
Number of rooms × $1(t \geq 2010)$		-0.02*
		(0.01)
Number of floors × $1(t \geq 2010)$		-0.03
		(0.08)
Wood × $1(t \geq 2010)$		0.39*
		(0.20)
Lightweight concrete × $1(t \geq 2010)$		0.05
		(0.09)
Outhouse × $1(t \geq 2010)$		0.02
		(0.04)
Car park × $1(t \geq 2010)$		0.09
		(0.19)
Car park double × $1(t \geq 2010)$		0.07
		(0.22)
Roof tile × $1(t \geq 2010)$		0.04
		(0.04)
Roof fibercement asbestos × $1(t \geq 2010)$		-0.09**
		(0.03)
Stove heating × $1(t \geq 2010)$		-0.43
		(0.38)
Heatpump heating × $1(t \geq 2010)$		-0.16
		(0.33)
Electric heating × $1(t \geq 2010)$		-0.30
		(0.32)
Central heating × $1(t \geq 2010)$		-0.30
		(0.32)
District heating × $1(t \geq 2010)$		-0.30
		(0.36)
Nearest town × $1(t \geq 2010)$		0.00***
		(0.00)
Urban diversity × $1(t \geq 2010)$		-0.00
		(0.00)
Proximity to forest × $1(t \geq 2010)$		0.00
		(0.00)
Proximity to lake × $1(t \geq 2010)$		0.00
		(0.00)
Proximity to wetland × $1(t \geq 2010)$		-0.00
		(0.00)
Proximity to coast × $1(t \geq 2010)$		0.00
		(0.00)
Proximity to main road × $1(t \geq 2010)$		-0.00**
		(0.00)
Proximity to train station × $1(t \geq 2010)$		-0.00
		(0.00)
Proximity to national railway × $1(t \geq 2010)$		-0.00
		(0.00)
Proximity to windmill × $1(t \geq 2010)$		0.00
		(0.00)
Constant	12.43***	12.54***
	(0.27)	(0.15)
Control Var	X	X
Area FE	X	X
Year FE	X	X
Changing implicit prices		X
Observations	5,905	5,905
R-squared	0.45	0.46

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Clustered std. errors in parenthesis. $dist_i$ is a constructed variable that measures the distance in kilometers to a public school as of before 2010 for all dwellings. For dwellings located in the treatment area after 2010, this variable measures the distance to the nearest public school closure.